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Dissipative phase transitions in systems with nonreciprocal effective interactions

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The reciprocity of effective interparticle forces can be violated in various open and nonequilibrium systems, in particular, in colloidal suspensions and complex (dusty) plasmas. Here, we obtain a criterion under which a nonreciprocal system can be strictly reduced to a pseudo-Hamiltonian system with a detailed dynamic equilibrium. In particular, the criterion is satisfied for catalytically active colloids interacting via nonreciprocal diffusiophoretic forces. However, in the general case, when this criterion is not satisfied, the steady state is determined by the interplay between dissipation and the energy source due to the nonreciprocity of interactions. The results indicate the realization of bistability and dissipative spinodal decomposition in a broad class of systems with nonreciprocal effective interactions.

1 Introduction

Open and nonequilibrium systems of various kinds are intrinsic to physics, chemistry, materials science, and even multi-agent systems and networks. Therefore, understanding of mechanisms responsible for collective phenomena in such systems, *e.g.*, dynamic self-organization, dissipative phase transitions, and critical phenomena, is important for various fundamental and applied problems.

The interaction nonreciprocity, *i.e.*, a violation of the action–reaction symmetry for effective interparticle interactions, is an intrinsic feature of open systems. The effective interactions in these systems are mediated by a nonequilibrium environment:¹ examples include colloidal suspensions and complex plasmas, where interactions are mediated by flows,^{2–8} nonequilibrium fluctuations,⁹ optical beams,¹⁰ diffusiophoresis,^{11–14} and plasma wakes.^{1,15–27} Another prominent mechanism of the nonreciprocity (which is, however, beyond the scope of the present paper) is due to relativistic effects, affecting the magnetic and retarded interactions in electrodynamics.^{28,29}

The interest in systems with nonreciprocal effective interactions is motivated by their abundance in nature, and their fundamental and applied importance, which can be easily illustrated. As a representative example of the plasma state of soft matter,¹⁷ one can consider two-dimensional (2D) complex (dusty) plasmas.^{1,8} In such systems, charged microparticles levitate in a weakly ionized gas, *e.g.*, in the plasma of a capacitively coupled radio-frequency discharge. Due to the interplay between

gravitational and electric forces, particles can form monolayered or multilayered structures, depending on the experimental conditions. The interactions between microparticles are usually affected by the vertical plasma flow, which generates the so-called ‘plasma wakes’ downstream from each particle.^{30–32} The effect of plasma wakes leads to nonreciprocity of effective interaction,³³ which depends on the plasma discharge parameters and can be tuned in experiments.²³ As a result, the microparticles can efficiently acquire energy from plasma flow, which opens exciting prospects for particle-resolved studies of nonequilibrium phenomena in systems with activation thermal behavior, in particular, propagation of flame fronts,²⁶ critical heat phenomena,³⁴ and thermoacoustic instability in chemically-reactive media.³⁵

Catalytically active colloidal suspensions^{12,36} represent another class of soft matter with nonreciprocal interactions. The particles interact due to chemical reactions occurring at their surface in a solvent.^{12,36} Each particle generates an inhomogeneous chemical field in the solvent. The diffusiophoretic force acting on a colloidal particle from another one is proportional to the diffusiophoretic mobility of the first particle and to the surface activity of the second one. The mobility and activity characterize a response of the particle to chemical gradients and how chemicals are produced (or consumed) at the surface of the particle, respectively. In the case of relative disparity of the parameters, the nonreciprocity of interactions arises.¹² From a practical point of view, along with micro-fluidic^{37,38} and electrically-induced interactions,^{39–44} catalytically active colloids provide a prospective way to construct long-range interactions for self-assembly of complex materials.^{45,46}

The nonreciprocity of interactions can be realized due to external forces, inducing flows which mediate the interactions, like in the case of complex plasmas, as well as due to chemical reactions, like in the case of catalytically active colloids. Active

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matter, bacteria, and self-propelled particles (microswimmers), whose dynamics is determined by hydrodynamic flows and a local chemical environment, can exhibit nonreciprocal effective interactions.^{47,48} The nonreciprocity of effective interactions plays an important role in dynamics of social (multi-agent) systems, including groups of animals,^{49–52} human crowds,⁵³ pedestrian systems,^{54–57} systems of unmanned vehicles,⁵⁸ and epidemic systems,⁵⁹ which can also be described by dynamic equations with effective nonreciprocal interactions.

In the present Article, we study how the nonreciprocity of effective interactions affects the collective dynamics of systems. We generalize the solution obtained in ref. 1 and derive a criterion, under which the system exhibits a detailed dynamic equilibrium of the subsystems with different temperatures (a pseudo-Hamiltonian behavior). Otherwise, the nonreciprocal effective interactions lead to heating of the system and the resulting state is determined by the balance between the energy source and dissipation. The balance method is developed for fast and accurate analysis of steady states in nonreciprocal systems. Finally, we demonstrate that, if the nonreciprocity parameter depends on the distance between particles, a set of steady states can arise leading to dissipative spinodal decomposition in nonreciprocal systems, reported here for the first time.

2 Results

2.1 Pseudo-Hamiltonian systems

In ref. 1, it was shown that a binary system with a radially independent (constant) nonreciprocity of interactions exhibits pseudo-Hamiltonian dynamics. Here, we generalize this problem for an arbitrary number of subsystems with nonreciprocal interactions between them. We are solely focused on a broad class of interactions depending on the relative particle positions (but not on their velocities).

We consider an ensemble of particles subdivided into different subsystems (denoted by index s), so that the pair interactions within each subsystem is reciprocal, whereas the interactions between particles belonging to different subsystems are nonreciprocal. The dynamics of individual particles is described by the equation

$$m_{is}\dot{\mathbf{v}}_{is} = \sum_{j \neq i} \mathbf{F}_{js, is} + \sum_{s' \neq s} \sum_j \mathbf{F}_{js', is}, \quad (1)$$

where m_{is} and $\mathbf{v}_{is} \equiv \dot{\mathbf{r}}_{is}$ are the mass and velocity of the i -th particle in the s -th subsystem, respectively, and $\mathbf{F}_{js', is}$ is the force acting on the is -th particle by the js' -th particle. The first term on the right-hand side of eqn (1) describes the total force acting on the is -th particle due to interactions within the s -th subsystem and the second term represents the total force due to nonreciprocal interactions with other subsystems s' .

Let the action and reaction act along the radius vector between the interacting particles, but have different magnitudes, as is illustrated in Fig. 1(a). Then, $\mathbf{F}_{js', is} = F_{js', is}(\mathbf{r}_{js'} - \mathbf{r}_{is})/|\mathbf{r}_{js'} - \mathbf{r}_{is}|$ and we can write

$$F_{js', is} = (1 + \Delta_{s's})F_{ji}^{(r)}, \quad (2)$$

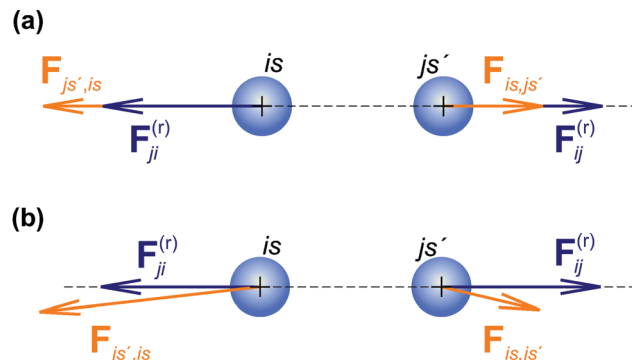


Fig. 1 Forces in a pair of nonreciprocally interacting particles. (a) Scalar nonreciprocity: the action and reaction act with different magnitudes along the radius vector between the particles. (b) Tensor nonreciprocity: the action and reaction have different magnitudes and are noncollinear.

$$F_{ji}^{(r)} = \frac{1}{2}(F_{js', is} + F_{is, js'}), \quad \Delta_{s's} = \frac{F_{js', is} - F_{is, js'}}{F_{js', is} + F_{is, js'}},$$

where $F_{ji}^{(r)}$ is the reciprocal part of the interaction and $\Delta_{s's} = -\Delta_{ss'}$ is the scalar nonreciprocity, characterizing the visible deviation from Newton's third law for effective interactions and generally depending on the distance $r_{ji} = |\mathbf{r}_j - \mathbf{r}_i|$ between particles. One can see that the forces $\mathbf{F}_{ji}^{(r)}$ obey Newton's third law for effective interactions and, hence, the potential energy $\phi(r_{ji})$ can be introduced so that $\mathbf{F}_{ji}^{(r)} = -\partial\phi(r_{ji})/\partial\mathbf{r}_i$.

In the same manner as in ref. 1, we can find the renormalizing constants α_s , which provide the detailed action-reaction symmetry between each pair of particles in subsystems s' and s , i.e.

$$\alpha_s F_{js', is} = \alpha_{s'} F_{is, js'}. \quad (3)$$

This approach is equivalent to renormalization of masses and interaction potentials in ref. 1.

One can see that, in the case of scalar radially-independent nonreciprocity, eqn (3) is satisfied, if the forces can be factorized as follows

$$F_{js', is} = f(p_{s'})g(p_s, p_{s'})F_{ji}^{(r)}, \quad (4)$$

where $f(p)$ is an arbitrary function of parameter p and $g(p_s, p_{s'}) = g(p_{s'}, p_s)$ is an arbitrary function, which is invariant relative to the transposition of variables. Here we introduced parameter p_s for subsystem s , which is related to nonreciprocal interactions: For instance, p_s can be a vertical coordinate of particle layers in the case of complex (dusty) plasmas¹ or the ratio k/μ of chemical activity k to phoretic mobility μ in catalytically active colloids.^{12,36}

Using eqn (4), one can obtain the solution of eqn (3):

$$\alpha_s = Af(p_s), \quad \Delta_{s's} = \frac{f(p_{s'}) - f(p_s)}{f(p_{s'}) + f(p_s)}, \quad (5)$$

where A is an arbitrary constant. This is a simple generalization of the solution found in ref. 1 for the binary system.

The solution means that, if the forces $F_{js', is}$ can be represented in the form (4), the system exhibits conservative (pseudo-Hamiltonian) dynamics and can be strictly described in terms

of equilibrium statistical mechanics. The constants α_s determine the temperatures of the subsystems, $T_s = \alpha_s \tilde{T}$, where $\tilde{T}$ is the pseudotemperature, corresponding to the renormalized Hamiltonian function (see Appendix A).

For instance, in the case of diffusiophoretic forces acting in catalytically-active colloids, one can write^{12,36}

$$F_{js',is} = p_{s'} \mu_s \mu_{s'} / r_{ji}^2, \quad (6)$$

where $p_s = k_s / \mu_s$. Using eqn (4)–(6), we have in this case $f(p_s) = p_s$, so that

$$A_{s's} = \frac{p_{s'} - p_s}{p_{s'} + p_s}, \quad \frac{T_{s'}}{T_s} = \frac{p_{s'}}{p_s}. \quad (7)$$

Solution (7) means that catalytically active colloidal suspensions can be reduced to pseudo-Hamiltonian systems, in spite of the broken action–reaction equality for diffusiophoretic interactions.[†] Owing to the nonreciprocity, catalytically active colloids tend to a state of detailed dynamical equilibrium with different temperatures for different subsystems, while the strong viscous damping ensures equilibration with the solvent.

The considered approach can be generalized for the tensor nonreciprocity of interactions, *i.e.*, for the case where the action and reaction have different magnitudes and are noncollinear, as is illustrated in Fig. 1(b), see Appendix B.

Summarizing the results, one can see that a system with nonreciprocal effective interactions can demonstrate conservative dynamics if the nonreciprocity $A_{s's}$ is scalar and can be represented by equation (5). When this criterion is not satisfied, the subsystems cannot remain in the detailed dynamic equilibrium, which leads to an increase in the kinetic energy of the system. As we demonstrate below, the interplay between the energy source and dissipation can result in rich scenarios of dynamic evolution of systems with nonreciprocal interactions.

We point out that dissipation plays an important role in the dynamics of systems with nonreciprocal effective interactions. The dissipation can be relatively weak, as in complex plasmas (where the individual particle dynamics is virtually undamped) or strong, like in colloidal suspensions (where the dynamics is governed by the overdamped Brownian regime).⁸ However, the principal aim of the analysis presented in this section is to identify and illustrate the essential difference between nonreciprocal systems where interactions can be described by a pseudo-Hamiltonian function (thus allowing the detailed dynamical equilibrium, as in the example above) and those where such presentation cannot be made (*i.e.*, essentially nonequilibrium systems). Therefore, in what follows we consider weakly dissipative systems and focus on the latter class of nonreciprocal systems, with $A_{s's}$ being either independent of the interparticle distance r (Section 2.2), or of $A_{s's}(r)$ (Section 2.3).

2.2 Radially-independent (constant) nonreciprocity

To illustrate departure from the detailed dynamical equilibrium, let us consider a model interaction with exponential nonreciprocity between the subsystems,

$$f(p_s) = \exp(-2\gamma p_s), \quad (8)$$

where γ is a constant characterizing the nonreciprocity and $p_s = s$ is the subsystem index. Substituting eqn (8) in eqn (5) and keeping in mind that $T_s/\alpha_s = \text{const}$, we obtain

$$A_{s's} = \tanh[\gamma(p_s - p_{s'})], \quad \frac{T_{s'}}{T_s} = e^{2\gamma(p_s - p_{s'})}. \quad (9)$$

Thus, the detailed equilibrium is realized when $A_{s's}$ obeys the hyperbolic tangent (HT) law.

We performed molecular dynamics (MD) simulations without a thermostat (for the details see Appendix C). As a representative example, we considered a system consisting of $N_s = 25$ subsystems with equal distribution of particle number in the subsystems and $\gamma = 0.03$. In this case, $p_s = s$ and $s = 1 \dots N_s$ in eqn (8) and (9). In addition to the HT law (9), we considered different $A_{s's}$, corresponding to the first, third, and fifth order expansions of the HT law:

$$A_{s's} = \gamma(s - s') - \frac{1}{3}\gamma^3(s - s')^3 + \frac{2}{15}\gamma^5(s - s')^5 + \dots \quad (10)$$

Note that, since the concept of temperature cannot be applied to a nonequilibrium state, we use an average kinetic energy per particle K in the system (or K_s for the energy per particle in the subsystem s). Temperature $T_s = \alpha_s \tilde{T}$ is used only for the pseudo-Hamiltonian case, where the average kinetic energy is equal to the pseudotemperature, $K = \tilde{T}$. Using eqn (9), we readily obtain the following expression for the ratio $T_s/\tilde{T}$:

$$\frac{T_s}{\tilde{T}} = \frac{N_s \alpha_s}{\sum_{s=1}^{N_s} \alpha_s} = \frac{N_s e^{(1+N_s)\gamma} \sinh \gamma}{\sinh \gamma N_s} \exp(-2\gamma s), \quad (11)$$

where N_s is the total number of the subsystems. Comparing eqn (9) and (11), one can see that $A = N_s e^{(1+N_s)\gamma} \sinh \gamma / \sinh \gamma N_s$.

Fig. 2(a) presents the MD results for time evolution of the kinetic energy at nonreciprocity corresponding to expansions of different orders indicated by eqn (10). One can see that even a weak deviation from the HT law (9) leads to heating of the system, since the forces cannot be represented in the form of (4). After the system is sufficiently warmed, $K(t)$ has the asymptotic behavior $K \propto t^{2/3}$.¹ The distribution of kinetic energies K_s in the subsystems is similar to the pseudo-Hamiltonian case since K_s/K is close to the analytical result (11), as is shown in the inset of Fig. 2(a). Note that, on the contrary to the radially-dependent nonreciprocity $A_{s's}(r)$ considered below, $K_{s'}/K_s$ does not depend on the energy K .

In the case of $A_{s's}$ determined by the HT law (9), the system becomes conservative (pseudo-Hamiltonian) as a whole but is dynamically decomposed to the subsystems with different temperatures T_s . The average kinetic energy $K = \tilde{T}$ does not change with time, as shown by brown symbols in Fig. 2(a). Using pseudotemperature $\tilde{T}$ and eqn (11), one can calculate the

[†] Strictly speaking, the diffusiophoretic forces can deviate from eqn (6) at short distances, which could be important for dense colloids. The detailed analysis of this effect will be performed in a separate work.

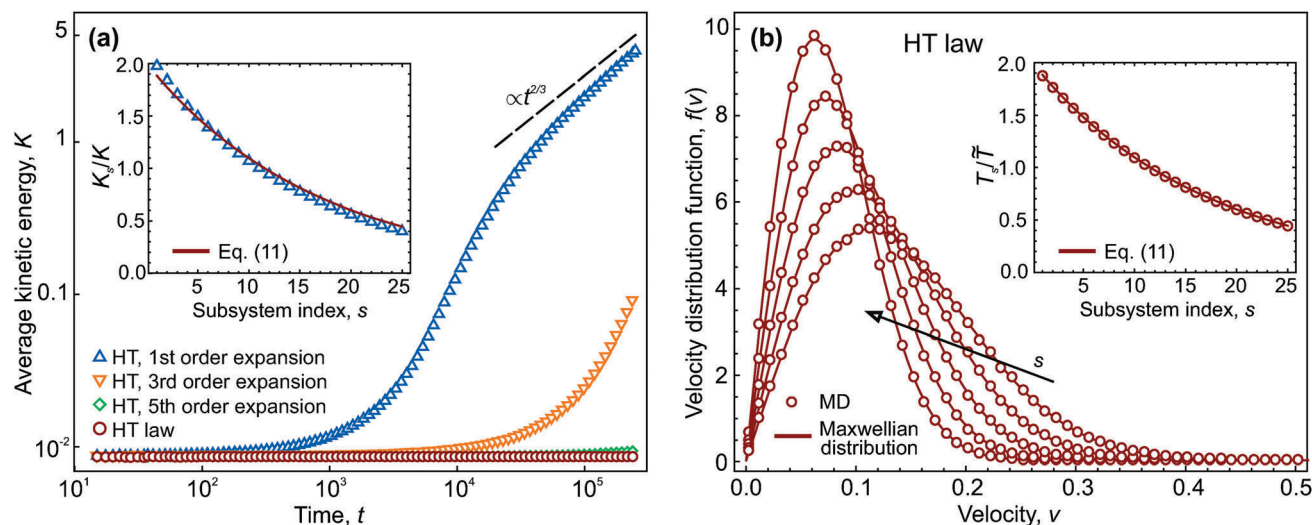


Fig. 2 Results for a system with the constant nonreciprocity of the interaction in the system with $m_{is} = 1$, $N_s = 25$. (a) Time evolution of the average kinetic energy per particle $K(t)$. Blue, orange, green, and brown symbols correspond to the Taylor expansions (10) to the first, third, and fifth orders, and hyperbolic tangent (HT) law (9), respectively. The inset presents the ratio K_s/K for the first order expansion of the HT law (symbols), the solid line corresponds to (11). (b) Velocity distribution functions in subsystems, in the case of the HT law (9). Symbols correspond to molecular dynamics (MD) results and solid lines correspond to the Maxwellian velocity distributions at temperatures T_s determined by eqn (11) for $s = 5, 10, 15, 20$, and 25 , increasing along the arrow. The inset presents the distribution of temperatures in the subsystems. Symbols correspond to MD results and the solid line corresponds to the analytical solution (11).

temperatures of the subsystems T_s . Fig. 2(b) demonstrates the velocity distributions for the subsystems, where the symbols correspond to MD results and solid lines correspond to the Maxwellian velocity distributions with analytically calculated temperatures (11), as shown in the inset of Fig. 2(b). Fig. 2(b) illustrates the correctness of our theoretical solution and it is in an excellent agreement with MD results.

2.3 Dissipative phase transitions

To study the effect of the radially-dependent nonreciprocity $\Delta_{s's}(r)$, we considered a binary system of repulsing particles with an equivalent number of particles in the subsystems in the Langevin thermostat, as a representative example. The reciprocal part of the interactions was the same as in Section 2.2, the nonreciprocity $\Delta_{s's}(r)$ is explained in Appendix C. The dynamics of this system without a thermostat has been considered in ref. 1, while the effect of the Langevin thermostat has not been studied before. In ref. 14, another kind of interaction (Yukawa potential) with the Brownian thermostat was used. We studied the binary system using MD simulations and the balance method for the analysis of the steady states, whose details are provided in Appendices C and D. The simulations and analysis were performed at different damping times τ and thermostat temperatures T_{th} .

Fig. 3 presents the results obtained at the thermostat temperature $T_{th} = 10^{-2}$. Evolution of energy $K(t)$ at different damping times τ of the thermostat is shown in Fig. 3(a) by the red and blue lines for the 'cold' and 'hot' initial states, respectively. Curve A corresponds to the non-dissipative case, where $\tau \rightarrow \infty$ and the system is heated irrespective of the initial state. However, with a decrease in τ , the evolution becomes determined

by the initial state, justifying the bistability of steady states, as is shown by dashed lines D–G in Fig. 3(a). Results for the asymptotic kinetic energy $K|_{t \rightarrow \infty} = K_\infty$ at different τ values are summarized in Fig. 3(b), where we can clearly see the bistability and hysteresis loop, which can be explained by analyzing the power of the energy source and dissipation.

The time evolution of the energy $K(t)$ is governed by the interplay between the powers of energy release P_{NR} and dissipation P_{damp} , owing to the nonreciprocity of interactions and dissipation.[‡] Fig. 3(c) presents the power P_{NR} (thick orange lines) and powers P_{damp} (light blue solid lines for the cases C and H and dashed lines for the cases D–G related to the hysteresis loop as shown in Fig. 3(b)). Since the kinetic energy K increases at $P_{NR} > P_{damp}$ and decreases otherwise, the possible steady states are determined by the balanced equation $P_{NR} = P_{damp}$. At large τ values, the states with high energies K_∞ are stable, as in case C. In the opposite limit of small τ values, the energy K_∞ tends to its asymptotic value of about T_{th} , as in case H. However, at τ values corresponding to cases E and F, points of tangency arise, as is shown by the light blue dashed lines in Fig. 3(c). In the τ range between these two points of tangency, the curves P_{damp} and P_{NR} intersect at three points instead of one. Thus, as a result of the interplay between the energy source and dissipation, the system has two steady states.

The steady states calculated using the balance method are shown in Fig. 3(b) by the black solid line. One can see that the method determines asymptotic steady states with a high accuracy.

[‡] The situation can be more complicated in strongly-coupled systems, where the dynamics of particles is determined by the derivatives of the interaction forces. The heating in this case is caused by the instability onset due to the inequality of the derivatives. A detailed analysis of this effect should be performed in future.

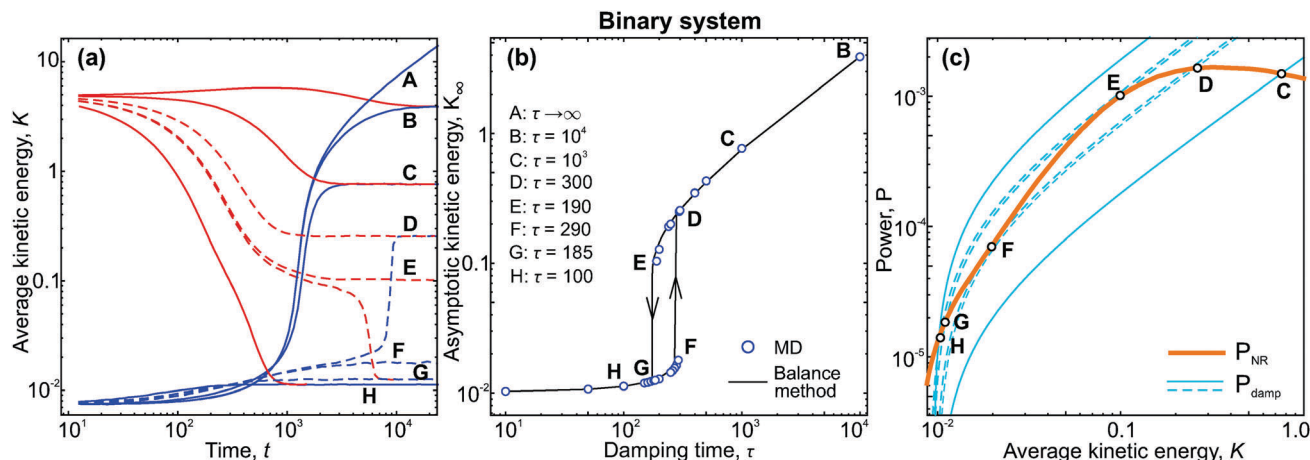


Fig. 3 Results for a binary system with radially-dependent nonreciprocity of interactions. (a) Time evolution of the average kinetic energy $K(t)$ at different damping times τ from the initial states with low and high energies, shown by red and blue lines, respectively. Dashed lines are related to the hysteresis loop shown in panel (b). (b) Asymptotic kinetic energy K_∞ at different damping times τ . Blue symbols represent MD results, the black solid line is obtained using the balance method explained in Appendix D. (c) Powers caused by nonreciprocity P_{NR} (orange thick lines) and interaction with the thermostat P_{damp} (light blue lines) for cases C and H (solid lines) and for cases D–G (dashed lines) versus the average kinetic energy (see Appendix D for details). The results are shown at $T_{th} = 10^{-2}$, and the damping times τ are notified in panel (b) and labeled by letters A–H. Masses of all particles are $m_{is} = 1$.

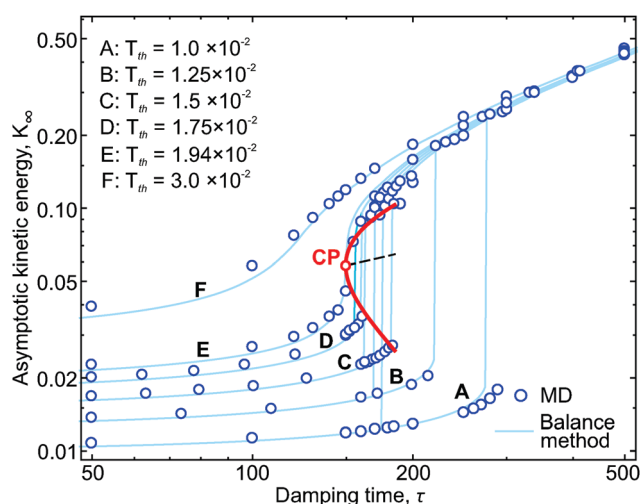


Fig. 4 Dissipative spinodal decomposition in the system with a non-reciprocal effective interaction. The asymptotic energy K_∞ versus the damping time τ at different thermostat temperatures T_{th} . Symbols represent MD results, and light blue solid lines are obtained using the balance method at different temperatures T_{th} marked by letters A–F. The dissipative spinodal decomposition (12) is shown by the red solid line with the critical point (CP) in the vertex, while the dashed black line is the averaged energy of activated and inactivated states.

However, the balance method requires much less computation time than MD simulations and subsequent analysis of $K(t)$.

The observed behavior of steady states with change in τ indicates the existence of dissipative phase transitions in the systems with nonreciprocal interactions between particles. To illustrate this, the steady states were calculated at different T_{th} values and the results are presented in Fig. 4. With an increase in the thermostat temperature T_{th} , energy dissipation and the width of the hysteresis loop decrease, and dynamic decomposition

vanishes for arbitrary τ values at the thermostat temperatures higher than the critical temperature, $T_{th} > T_{CP}$. Lines corresponding to different thermostat temperatures T_{th} play the role of dissipative ‘isotherms’ as presented in Fig. 4 in the τ – K coordinates. One can see that the limit points on the isotherms form a line of dissipative spinodal decomposition that has the vertex at the dissipative critical point (CP) and is shown by the red solid line. Note that the dissipative spinodal decomposition and the critical point are considered here in the context of dynamic decomposition and bifurcation behavior of the system at which the activated and inactivated states can exist.

We analyzed the dissipative spinodal decomposition near the critical point and observed that it can be described approximately by the power law

$$K_+ - K_- \propto (\tau - \tau_{CP})^\beta, \quad (12)$$

where τ_{CP} is the critical value of the damping time, and $K_\pm$ are the kinetic energies of activated and inactivated steady states denoted by $\pm$ indices, respectively. The parameters of the critical point (τ_{CP} , K_{CP} , and T_{CP}) are determined by a particular form of the nonreciprocal interaction. The average kinetic energy exhibits linear dependence with $\tau - \tau_{CP}$ near the critical point.

For results shown in Fig. 4, we observed that $\beta \simeq 1/2$, which is close to the mean-field type of the critical behavior. Possibly, this is due to the fact that the dissipative phase transition occurs simultaneously in the whole system, providing similarity to the long-range interactions and related critical behavior.⁶⁰ Note also that there is no dissipative ‘binodal’ line in this system in principle since the system cannot coexist simultaneously in a steady state with two different K values. Otherwise, it would require the coexistence of two different effective temperatures in each of the subsystems, where this is impossible since the interactions are reciprocal. A more detailed study of the

dissipative critical behavior in systems with nonreciprocal interactions, including the calculation of critical exponents, should be performed in future.

2.4 Discussion

The obtained results allow us to better understand a variety of experimentally observed nonequilibrium phenomena in complex (dusty) plasmas.^{20,21,26,34,35,61–64} In spite of a more complicated character of the wake-mediated interactions in these systems,^{§ 23} their behavior can be understood in terms of dissipative phase transitions and interplay between the energy release and dissipation. To illustrate this, below we discuss (i) crystallization, (ii) dissipative hysteretic behavior, (iii) thermal activation, (iv) fronts of nonequilibrium melting, and (v) thermoacoustic instability.

Typical experimental regimes of crystallization in complex plasmas can be readily explained with Fig. 3(b). If particles are introduced in the plasma discharge at a sufficiently low pressure, they often do not form a crystal. The resulting steady state of the system corresponds to a high-temperature (activated) state (C or D in Fig. 3(b)). Crystallization can be induced by increasing the damping rate of particles, controlled by the gas pressure. Then the system undergoes a dissipative transition from a “hot” to “cold” state, and thus crystallizes. After reducing the pressure back to the initial value, the crystal may remain stable (depending on conditions, in particular on the number of particles). This represents the dissipative hysteretic behavior illustrated in Fig. 3(b).

Low- and high-temperature steady states can be observed in experiments. Fig. 3(b) shows that, in the region of bistability, the two branches in the hysteresis loop are separated by the energy gap. This is indicative of the thermal activation. As one can see from Fig. 3(c), the activation energy is the difference between the energy corresponding to the unstable solution of the balance equation $P_{\text{NR}}(K) = P_{\text{damp}}(K)$ (where $dP_{\text{NR}}/dK > dP_{\text{damp}}/dK$) and the energy of the low-temperature state.

The thermal activation has been systematically investigated in experiments with nonequilibrium melting of plasma crystals.^{26,34,35,61} “Flame fronts” studied in ref. 26 are the fronts of self-sustained melting due to activation into the (high-temperature) fluid regime. This kind of nonequilibrium melting has a direct physical analogy to the flame fronts in chemically reactive solids,²⁶ characterized by an activated heat release, the two-zone structure of the self-similar temperature profile, and temperature saturation behind the front. In terms of dissipative phase transitions discussed in the present paper, the flame fronts are the spatially-propagating waves of transition to the activated (high-temperature) regime, accompanied by crystal melting.

Fig. 3(c) demonstrates that, for a certain temperature range, the nonreciprocity provides a positive thermal feedback: The

resulting energy release in a fluid complex plasma increases with the number density and the kinetic energy of particles. Since the medium is heated during the compression phase and cooled during rarefaction, this may result in thermoacoustic instability, which has been reported recently in ref. 35.

We point out that the low- and high-temperature steady states are separated due to the dissipative spinodal decomposition, as shown in Fig. 4. In the context of the above-discussed flame fronts,²⁶ this peculiarity may lead to rapidly changing temperature profiles across the fronts. However, if the thermostat temperature exceeds the critical value, the kinetic energy of particles can be varied continuously, by changing the damping rate. In complex plasmas, the effective thermostat temperature can be controlled, *e.g.*, by means of rapidly moving laser beams.⁶⁸ A combination of tunable thermostat temperature and damping rate opens perspectives for studies on collective fluctuations^{69–76} and activation phenomena in fluid complex plasmas, which should be explored in future studies.

3 Conclusions

In conclusion, we have studied the dynamics of systems with nonreciprocal effective interactions between particles. We have obtained the criterion for nonreciprocal forces, under which the system exhibits a detailed dynamic equilibrium and a pseudo-Hamiltonian can be introduced. In particular, catalytically active colloids can be described using the pseudo-Hamiltonian approach. The deviation of the nonreciprocity from the derived criterion leads to heating of the system. In the case of constant nonreciprocity, the steady state is determined uniquely by the balance between the energy source owing to nonreciprocity and dissipation due to damping. However, in the case of radially-dependent nonreciprocity, bistability can be realized owing to a complicated dependence of powers of the energy source and dissipation on the average energy of the system. At a fixed thermostat temperature, steady states have a hysteretic behavior, which indicates the existence of dissipative phase transitions occurring with the variation of the damping time in the range between two bifurcation points. The width of the hysteresis loop decreases with an increase in the thermostat temperature and vanishes at parameters corresponding to the dissipative critical point. The choice of the toy interaction model is motivated by its simplicity and the ability to reveal generic effects induced by nonreciprocal interactions. The effects owing to long-range character of interactions, interplay between softness of repulsion and attraction, and anisotropy of the interactions are worth exploring in future studies.

The results open novel prospects for the analysis of phenomena in systems where Newton's third law is broken for effective interactions between particles since a deviation from eqn (5) for nonreciprocity can play the role of a small parameter for prospective theories. If nonreciprocal forces can be written in the form of (4), the behavior of these nonequilibrium systems can be studied strictly within equilibrium statistical mechanics.

§ The wake-mediated interactions in complex plasmas are characterized by a radially-dependent tensor nonreciprocity. In addition to the effects induced by a scalar nonreciprocity, this can cause a coupling between the in-plane and out-of-plane wave modes, leading to the mode-coupling instability in the crystalline^{63,65,66} and fluid^{64,67} regimes.

The physics of observed phase transitions is related to the nonlinear dependence between the power of the energy source and dissipation on the average kinetic energy in the system, which results in multiple steady states. Such a behavior is intrinsic to a wide range of dissipative systems of different natures, *e.g.*, in driven-dissipative superfluids,⁷⁷ open atomic many-body systems,^{78–81} neural networks,^{82–84} dissipative photonic systems,^{85–88} and complex (dusty) plasmas.⁸⁹

The results of this work reveal the role of the nonreciprocity of effective interactions and pave the way for prospective studies of various problems, including the analysis of dissipative phase transitions and critical behavior of systems with different nonreciprocal effective interactions; interplay between dissipative and thermodynamics phase transitions; and effects of nonreciprocity on collective fluctuations in strongly-coupled systems, *e.g.*, fluid, crystalline, and glassy states. We believe that our results will be useful for interdisciplinary areas of physics, chemistry, biology, and statistical mechanics of non-equilibrium soft matter.

Conflicts of interest

There are no conflicts to declare.

Appendix

A Pseudo-Hamiltonian

Here we show that, if the condition (4) for the nonreciprocal forces is satisfied (and, therefore, the parameters α_s and $\Delta_{s's}$ are determined by eqn (5)), one can introduce the pseudo-Hamiltonian which is conserved in the system with nonreciprocal interactions.

One can introduce the following energy (Hamiltonian) H_s for the s -th subsystem:

$$H_s = \sum_i \frac{1}{2} m_{is} \mathbf{v}_{is}^2 + \sum_{i < j} \varphi(\mathbf{r}_{js} - \mathbf{r}_{is}). \quad (13)$$

Using eqn (1) and (2), we readily obtain that the time derivative of H_s is determined by the total powers of the works performed by other subsystems s' on the s -th subsystem (since the interactions inside each subsystem are reciprocal and do not contribute to a change of the energy):

$$\frac{d}{dt} H_s = \sum_{s' \neq s} \sum_{j,i} (1 + \Delta_{s's}) \mathbf{v}_{is} \cdot \mathbf{F}_{js',is}^{(r)}. \quad (14)$$

The (nonreciprocal) interactions between different subsystems are nonpotential and cannot be included in the Hamiltonian $H_{s'}$,¹ but their effect is accounted on the right side of eqn (14). One can see that if $\Delta_{s's} \neq 0$, the uncompensated energy is released because of the nonreciprocal interaction between the subsystems.

Consider the following linear combination of the Hamiltonians H_s in a system where $\Delta_{s's}$ does not depend on r_{ji} :

$$\mathcal{H} = \sum_s \alpha_s H_s, \quad (15)$$

where α_s is the renormalizing coefficient for subsystem s , introduced in eqn (3).

Using eqn (13)–(15), we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{H} &= \sum_{s' \neq s} \sum_{j,i} \alpha_s (1 + \Delta_{s's}) \mathbf{v}_{is} \cdot \mathbf{F}_{js',is}^{(r)} \\ &= \sum_{s < s'} [\alpha_s (1 + \Delta_{s's}) P_{s's} + \alpha_{s'} (1 + \Delta_{ss'}) P_{ss'}] \\ &= \sum_{s < s'} \alpha_s (1 + \Delta_{s's}) (P_{s's} + P_{ss'}), \end{aligned} \quad (16)$$

where $P_{s's} = \sum_{i,j} \mathbf{v}_{is} \cdot \mathbf{F}_{js',is}^{(r)}$.

Since $\mathbf{F}_{ji}^{(r)} = -\partial \varphi(\mathbf{r}_j - \mathbf{r}_i) / \partial \mathbf{r}_i$, we obtain

$$\begin{aligned} P_{s's} + P_{ss'} &= \sum_{i,j} (\mathbf{v}_{is} - \mathbf{v}_{js'}) \cdot \mathbf{F}_{js',is}^{(r)} \\ &= -\frac{d}{dt} \sum_{i,j} \varphi(\mathbf{r}_{js'} - \mathbf{r}_{is}) \\ &= -\frac{d}{dt} \varphi_{s's}, \end{aligned} \quad (17)$$

where we have introduced $\varphi_{s's} = \sum_{i,j} \varphi(\mathbf{r}_{js'} - \mathbf{r}_{is})$, which denotes the energy of the reciprocal part of the interaction between the subsystems s' and s .

Comparing eqn (16) and (17), one can see that our system is described by the pseudo-Hamiltonian

$$\tilde{H} = \sum_s \alpha_s H_s + \sum_{s < s'} \alpha_s (1 + \Delta_{s's}) \varphi_{s's}, \quad (18)$$

where the first and the second sums are related to the renormalized total energies of the subsystems and to the energy of the interactions between the subsystems, respectively. The pseudo-Hamiltonian (18) does not change with time, $d\tilde{H}/dt = 0$.

B Tensor nonreciprocity

In the case of tensor nonreciprocity illustrated in Fig. 1(b), the scalar nonreciprocity in eqn (2) should be replaced by the tensor,

$$1 + \Delta_{s's} \rightarrow \delta_{\mu\nu} + (\tilde{\Delta}_{s's})_{\mu\nu}, \quad (19)$$

where $\delta_{\mu\nu}$ is the Kronecker tensor (μ and ν are the tensor indices), and $(\tilde{\Delta}_{s's})_{\mu\nu}$ is the tensor of nonreciprocity introduced so that

$$\tilde{\Delta}_{s's} \mathbf{F}_{ji}^{(r)} = \frac{1}{2} (\mathbf{F}_{js',is} + \mathbf{F}_{is,j's'}). \quad (20)$$

Using eqn (2), we can represent the condition (3) in the form of

$$\alpha_s (1 + \tilde{\Delta}_{s's}) = \alpha_{s'} (1 + \tilde{\Delta}_{ss'}). \quad (21)$$

In the same manner as for the scalar nonreciprocity, we obtain the functional equation for the components

$$\alpha_s [\delta_{\mu\nu} + (\tilde{\Delta}_{s's})_{\mu\nu}] = \alpha_{s'} [\delta_{\mu\nu} + (\tilde{\Delta}_{ss'})_{\mu\nu}]. \quad (22)$$

For the diagonal components, $\mu = \nu$, eqn (22) coincides with eqn (3), and the solution is given by eqn (5), while for $\nu \neq \mu$, we have

$$\alpha_s(\tilde{A}_{s's})_{\mu\nu} = \alpha_{s'}(\tilde{A}_{ss'})_{\mu\nu} \quad (23)$$

Taking into account that $(\tilde{A}_{s's})_{\mu\nu} = -(\tilde{A}_{ss'})_{\mu\nu}$, eqn (23) is compatible with eqn (5) only if $(\tilde{A}_{s's})_{\mu\nu} = 0$ at $\mu \neq \nu$. This means that $(\tilde{A}_{s's})_{\mu\nu} = \delta_{\mu\nu}\tilde{A}_{s's}$ and the tensor nonreciprocity reduces to the scalar case (5).

C Details of MD simulations

Molecular dynamics simulations were performed for two-dimensional systems consisting of $N = 2 \times 10^4$ particles with the equal masses $m_{is} = m$ evenly distributed to $N_s = 25$ ($N_s = 2$) subsystems with a random uniform distribution of particles in the case of constant (variable) nonreciprocity.

The reciprocal interaction force was chosen in the same model form as in ref. 1, $\mathbf{F}_{ji}^{(r)} = -\partial\varphi(r_{ji})/\partial\mathbf{r}_i$, where

$$\varphi(r) = \begin{cases} \varepsilon(1 - r/r_0)^2/2, & r < r_0; \\ 0 & r \geq r_0; \end{cases} \quad (24)$$

where ε and r_0 are the characteristic magnitude and spatial scale of the interaction, respectively. Scalar radially-dependent nonreciprocity $\tilde{A}_{s's}(r)$ in our simulations was taken in the form of

$$\tilde{A}_{s's}(r) = (1 - r/r_0) \text{sign}(s' - s). \quad (25)$$

The performed simulations in the *NV* (number of particles and volume (area) of the system are constant) and *NVT* (the number of particles, volume, and temperature are constant) ensembles were based on their implementation in the LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator) package.^{90,91} Note that, on the contrary to the *NVE* ensemble (the number of particles, volume, and the total energy of the system are constant), the energy is not conserved in the *NV* ensemble owing to the nonreciprocity of interactions. Periodic boundary conditions were used to study the systems with an area density of $n = N/V = 0.8r_0^{-2}$.

The motion of the i -th particle in the presence of the Langevin thermostat (used in our simulations in the *NVT* ensemble) is described by the equation

$$m\ddot{\mathbf{r}}_i = \mathbf{F}_i - m\tau^{-1}\dot{\mathbf{r}}_i + \sqrt{2mT_{\text{th}}/\tau} \mathbf{R}_i(t), \quad (26)$$

where τ is the damping time, $\mathbf{F}_i = \sum_{j \neq i} \mathbf{F}_{ji}$ is the resulting force acting on the i -th particle, T_{th} is the thermostat temperature, and $\mathbf{R}_i(t)$ is the stochastic force acting on the i -th particle. The spatial components of this stochastic force constitute a Gaussian process satisfying the conditions

$$\langle \mathbf{R}_i(t) \rangle = 0, \quad \langle R_{i\alpha}(t)R_{j\beta}(t') \rangle = \delta_{\alpha\beta}\delta_{ij}\delta(t - t'), \quad (27)$$

where α, β are the spatial projection indices, i, j are the indices of different particles, and $\delta(t)$ denotes the Dirac delta function. A numerical time step of $\Delta t = 5 \times 10^{-3} \sqrt{mr_0^2/\varepsilon}$ was chosen. The numbers of time steps in various simulations were varied from 4×10^6 to 2×10^8 . The length, energy, and time were

represented in units of r_0 , ε , and $\sqrt{mr_0^2/\varepsilon}$, respectively, which corresponds to the calculations with r_0 , ε , and m equal to unity.

To show that the chosen time step is appropriate for simulating both low- and high-energy states of our system, we performed an additional run for energy state B (see Fig. 3) with the time step of $\Delta t' = 5 \times 10^{-4} \sqrt{mr_0^2/\varepsilon}$. The resulting average kinetic energy of $K = 3.91$ is very close to $K = 3.89$, the value obtained for the longer time step (with the relative error of about 0.5%).

D Balance approach to steady states

The balance method allows us to analyze steady states by comparing the powers of the energy source and dissipation in the system with nonreciprocal interactions. Here, we consider the binary system with an equivalent number of particles in the subsystems, but the approach can be generalized to an arbitrary number of subsystems.

In the simulations, we observed that, in the presence of a thermostat, both subsystems in a steady state have almost Maxwellian velocity distributions with some effective temperatures (average energies) of the subsystems, T_1 and T_2 . In the *NV* ensemble at $\tau \rightarrow \infty$, the energy K in the system grows and the velocity distributions in subsystems are non-Maxwellian, but the ratio K_1/K_2 at a given average energy of the system K is very close to that obtained in the presence of the thermostat, *i.e.*, to T_1/T_2 , if $\min\{T_1, T_2\} > T_{\text{th}}$. The examples of velocity distributions in the subsystems during heating are shown in Fig. 5. The lines represent the Maxwellian velocity distributions in the 2D system

$$f(v, T) = \frac{mv}{T} \exp\left(-\frac{mv^2}{2T}\right). \quad (28)$$

The parameters of a steady state obey the balance equation

$$N_1P_1 + N_2P_2 - (N_1 + N_2)P_{\text{NR}} = 0, \quad (29)$$

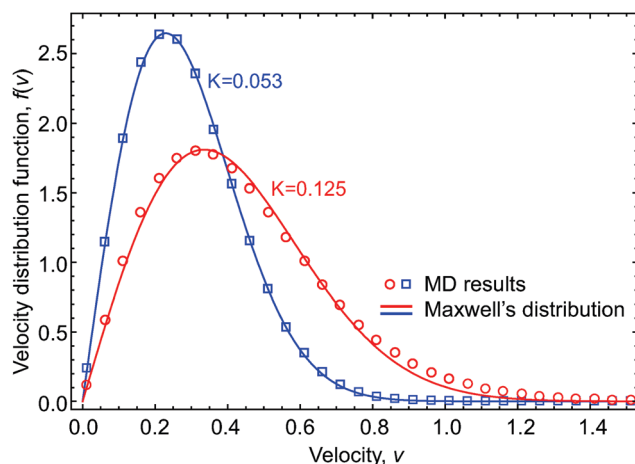


Fig. 5 Velocity distribution functions at the time of heating. Examples of velocity distribution functions in subsystems obtained using MD simulations (symbols) in comparison with the Maxwellian velocity distribution at the effective temperature determined by the average energies $T \simeq K$ in the subsystems (indicated near the curves).

where $N_{1,2}$ represents the number of particles in the subsystems, $P_{1,2} = 2\tau^{-1}(T_{1,2} - T_{\text{th}})$ is the energy dissipation power per particle in subsystems 1 and 2, and P_{NR} is the power of energy release per particle owing to the nonreciprocity of interactions. The damping power per particle in the system is $P_{\text{damp}} = (N_1 P_1 + N_2 P_2)/(N_1 + N_2)$.

To calculate $P_{\text{NR}}(K)$, the almost Maxwellian velocity distribution function (28) and different effective temperatures in the subsystems should be properly taken into account. This can be done in the following manner. At the first step, the dependence of K_1/K_2 on energy K is calculated using simulation in the NV ensemble at $\tau \rightarrow \infty$. Then, MD simulation is performed in the NV ensemble, where the subsystems are thermostated separately at temperatures T_1 and T_2 with the ratio $T_1/T_2 = K_1/K_2$ at a given average energy per particle K in the system. Finally, the energy source power can be calculated as $P_{\text{NR}} = dH/dt$, where $H = \sum_s H_s$ is the sum of the energies in the subsystems (13),

$$P_{\text{NR}} = \langle \mathbf{v} \cdot \mathbf{F} \rangle. \quad (30)$$

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